



CIVICPULSE

"Turn citizen reports into coordinated action."

Problem Statement

ID: SIH26043

A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships.

Theme: Disaster Management

Category: Software

Team Xieron

Team Leader: Mandhir Sawarn

TEKATHON 5.0 / SIH 2026 Submission

CivicPulse — From Citizen Report to Coordinated Resolution

The Problem

Citizens encounter civic and societal problems daily, but reporting, prioritization, routing, and follow-up can be highly fragmented and opaque.

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<h3>Why CivicPulse?</h3>
<p style="font-size: 18px;"><strong>Not just complaint registration.</strong><br>Transforms scattered reports into structured, prioritized civic challenges.</p>

<h3>Key Differentiators</h3>
<ul style="font-size: 16px; line-height: 1.4;">
  <li><strong>AI-assisted analysis:</strong> Severity & classification</li>
  <li><strong>Geographic intelligence:</strong> Hotspot & cluster detection</li>
  <li><strong>Department routing:</strong> Automated assignments</li>
  <li><strong>Resolution verification:</strong> Citizen loop closure</li>
</ul>
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Resolution Workflow



Technical Approach

Implementation Stack

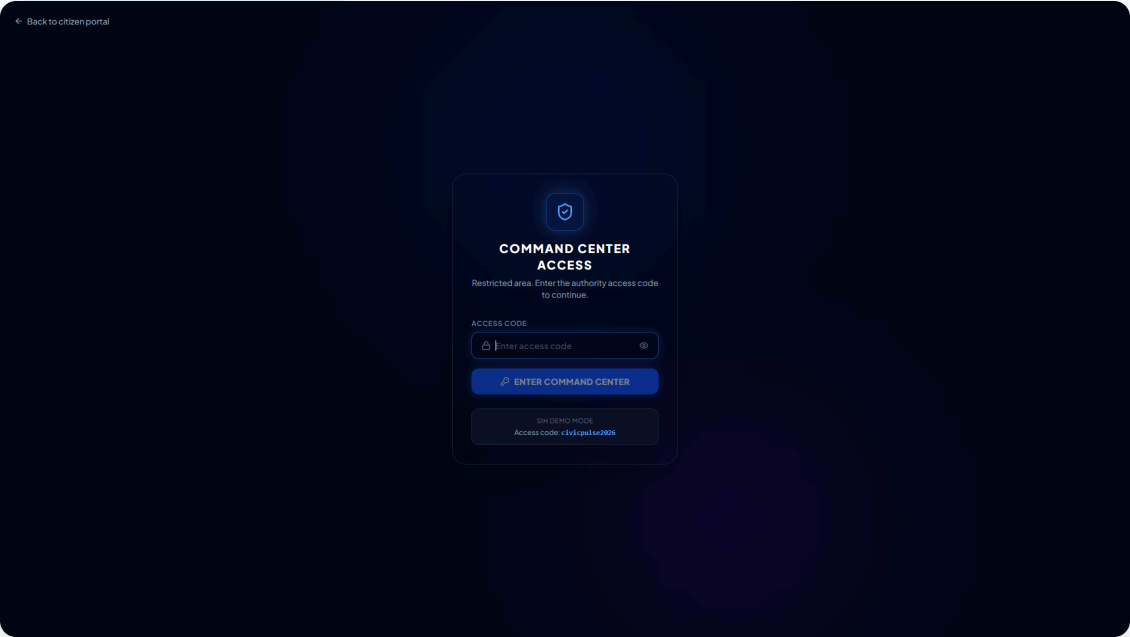
Built on a modern, modular, and extremely fast web architecture.

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<div style="display: flex; flex-wrap: wrap; gap: 8px; margin-bottom: 20px;">
  <span class="badge">React 19</span>
  <span class="badge">TypeScript</span>
  <span class="badge">Vite</span>
  <span class="badge">Tailwind CSS</span>
  <span class="badge">Zustand</span>
  <span class="badge">React Leaflet</span>
  <span class="badge">Recharts</span>
  <span class="badge">Framer Motion</span>
</div>

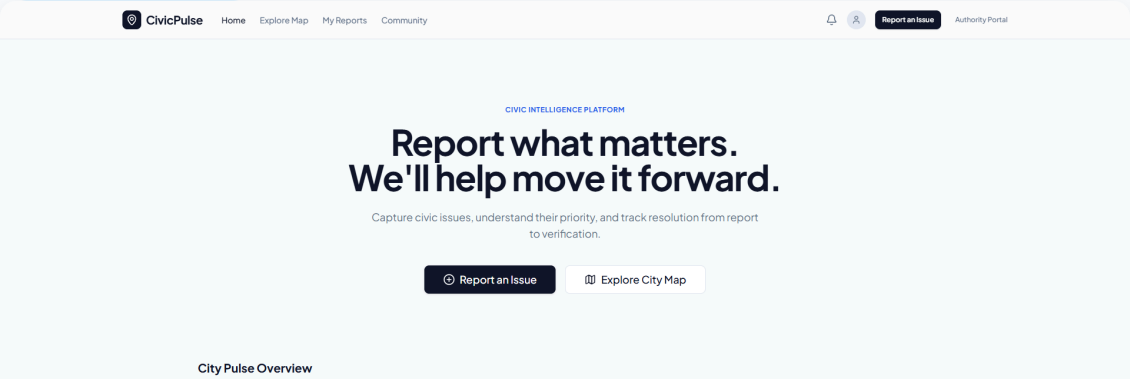
<h3>Architecture & AI Layer</h3>
<div style="font-size: 14px; background: white; padding: 15px; border-radius: 8px; border: 1px solid #e2e8f0; margin-bottom: 10px;">
  <strong>CITIZEN (React)</strong> <span class="arrow"></span> <strong>API (Zustand Store)</strong> <span class="arrow"></span> <strong>GIS (Leaflet)</strong> <span class="arrow"></span> <strong>AUTHORITY (React)</strong>
</div>

<ul style="font-size: 16px;">
  <li><strong>AI Layer</strong> Simulates issue classification and severity assessment.</li>
  <li><strong>GIS Layer</strong> Real-time geographic clustering and field team routing.</li>
</ul>
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GIS COMMAND CENTER



CITIZEN PORTAL



Feasibility & Viability

Technical Feasibility

- **Web-based architecture:** No installation required, high accessibility.
- **Modular frontend:** Clean separation of Citizen and Admin portals.
- **GIS Integration:** OpenStreetMap for zero-cost scalability.

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<h3>Operational & Scalability</h3>
<ul style="font-size: 16px;">
  <li><strong>Workflow:</strong> Seamless handoff from Citizen → Authority → Field Team.</li>
  <li><strong>Expansion Path:</strong> City <span class="arrow">→</span> Zone <span class="arrow">→</span> Ward <span class="arrow">→</span> Multiple Municipalities.</li>
</ul>
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Risks & Mitigation

Risk	Mitigation
Duplicate reports	AI similarity + geographic clustering logic
Incorrect AI categorization	Human review / authority override capability
False / low-quality reports	Photo evidence + moderation + verification
Location inaccuracies	Device GPS + manual map correction
AI hallucination errors	Human-in-the-loop validation checkpoints

Impact & Benefits



"CivicPulse creates a shared intelligence layer between citizens, authorities and solution partners."

Research, References & Prototype

Research & References

- **Problem Statement SIH26043:** Ministry directives on digital crowdsourcing.
- **Smart City Frameworks:** Municipal digital governance best practices and open-data standards.
- **Spatial Intelligence:** Academic research on GIS clustering and geographic hotspot detection for urban issues.
- **AI in Civic Tech:** Methodologies for automated severity assessment and image-based issue classification.
- **Participatory Design:** Studies on citizen engagement, loop-closure, and crowdsourcing platforms.

PROTOTYPE DEMO

Watch the working CivicPulse prototype

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<a href="https://youtu.be/qe4P2yn72H" style="display: inline-block; background: #f44444; color: white; padding: 5px 30px; border-radius: 30px; font-weight: 800; font-size: 20px; text-decoration: none; box-shadow: 0 4px 5px rgba(239, 68, 68, 0.4);">▶ WATCH PROTOTYPE VIDEO</a>
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